

A Practical Guide to Bayesian network

TOPIC -- BAYESIAN NETWORK

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)}$$

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Further reading Conrady S, Jouffe L (2015-07-01). Bayesian Networks and BayesiaLab – A practical introduction for researchers. Franklin, Tennessee: Bayesian USA. ISBN 978-0-9965333-0-0. Charniak E (Winter 1991). "Bayesian networks without tears" (PDF). AI Magazine. Kruse R, Borgelt C, Klawonn F, Moewes C, Steinbrecher M, Held P (2013). Computational Intelligence A Methodological Introduction. London: Springer-Verlag. ISBN 978-1-4471-5012-1. Borgelt C, Steinbrecher M, Kruse R (2009). Graphical Models – Representations for Learning, Reasoning and Data Mining (Second ed.). Chichester: Wiley. ISBN 978-0-470-74956-2.

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$$P(X_1 = x_1, \dots, X_n = x_n) = \prod_{v=1}^n P(X_v = x_v \mid X_j = x_j \text{ for each } X_j \text{ that is a parent of } X_v)$$

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$$\Pr(Y, Z \mid \text{do}(x)) = \Pr(Y, Z, X = x) \Pr(X = x \mid Z)$$

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$$P(X_v = x_v \mid X_j = x_j \text{ for each } X_j \text{ that is a parent of } X_v) = \prod_{i=1}^n P(X_v = x_v \mid X_i = x_i \text{ for each } X_i \text{ that is not a descendant of } X_v) \prod_{j=1}^m P(X_v = x_v \mid X_j = x_j \text{ for each } X_j \text{ that is a parent of } X_v)$$

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Local Markov property X is a Bayesian network with respect to G if it satisfies the local Markov property: each variable is conditionally independent of its

non-descendants given its parent variables:

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Markov blanket The Markov blanket of a node is the set of nodes consisting of its parents, its children, and any other parents of its children. The Markov blanket renders the node independent of the rest of the network; the joint distribution of the variables in the Markov blanket of a node is sufficient knowledge for calculating the distribution of the node. X is a Bayesian network with respect to G if every node is conditionally independent of all other nodes in the network, given its Markov blanket.

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$$\Pr(R = T \mid G = T) = \frac{\Pr(G = T, R = T)}{\Pr(G = T)} = \frac{\sum_{x \in \{T, F\}} \Pr(G = T, S = x, R = T)}{\sum_{x, y \in \{T, F\}} \Pr(G = T, S = x, R = y)}$$

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Several equivalent definitions of a Bayesian network have been offered. For the following, let $G = (V, E)$ be a directed acyclic graph (DAG) and let $X = (X_v), v \in V$ be a set of random variables indexed by V .

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To answer an interventional question, such as "What is the probability that it would rain, given that we wet the grass?" the answer is governed by the post-intervention joint distribution function

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A back-door path is one that ends with an arrow into X . Sets that satisfy the back-door criterion are called "sufficient" or "admissible." For example, the set $Z = R$ is admissible for predicting the effect of $S = T$ on G , because R d-separates the (only) back-door path $S \leftarrow R \rightarrow G$. However, if S is not observed, no other set d-separates this path and the effect of turning the sprinkler on ($S = T$) on the grass (G) cannot be predicted from passive observations. In that case $P(G \mid \text{do}(S = T))$ is not "identified". This reflects the fact that, lacking interventional data, the observed dependence between S and G is due to a causal connection or is spurious (apparent dependence arising from a common cause, R). (see Simpson's paradox) To determine whether a causal relation is identified from an arbitrary Bayesian network with unobserved variables, one can use the three rules of "do-calculus" and test whether all do terms can be removed from the

expression of that relation, thus confirming that the desired quantity is estimable from frequency data. Using a Bayesian network can save considerable amounts of memory over exhaustive probability tables, if the dependencies in the joint distribution are sparse. For example, a naive way of storing the conditional probabilities of 10 two-valued variables as a table requires storage space for $2^{10} = 1024$ values. If no variable's local distribution depends on more than three parent variables, the Bayesian network representation stores at most $10 \cdot 2^3 = 80$ values. One advantage of Bayesian networks is that it is intuitively easier for a human to understand (a sparse set of) direct dependencies and local distributions than complete joint distributions.

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P contains (but does not need to be entirely) a directed chain, $u \cdots \leftarrow m \leftarrow \cdots v$ or $u \cdots \rightarrow m \rightarrow \cdots v$, such that the middle node m is in Z , P contains a fork, $u \cdots \leftarrow m \rightarrow \cdots v$, such that the middle node m is in Z , or P contains an inverted fork (or collider), $u \cdots \rightarrow m \leftarrow \cdots v$, such that the middle node m is not in Z and no descendant of m is in Z . The nodes u and v are d-separated by Z if all trails between them are d-separated. If u and v are not d-separated, they are d-connected. X is a Bayesian network with respect to G if, for any two nodes u, v :

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with the term $\Pr(S = T \mid R)$ removed, showing that the action affects the grass but not the rain. These predictions may not be feasible given unobserved variables, as in most policy evaluation problems. The effect of the action $\text{do}(x)$ can still be predicted, however, whenever the back-door criterion is satisfied. It states that, if a set Z of nodes can be observed that d-separates (or blocks) all back-door paths from X to Y then